

Design, Simulation, and Characterization of Y-Branch and Directional-Coupler Silicon Photonic Mach–Zehnder Interferometers

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Abstract: This paper presents the end-to-end design, numerical simulation, fabrication, and optical characterization of two imbalanced Mach–Zehnder interferometers (MZIs) realized on a silicon-on-insulator (SOI) platform operating in the transverse-electric (TE) polarization at a central wavelength of 1550 nm. Both interferometers employ an identical 500×220 nm single-mode strip waveguide but incorporate distinct power-splitting mechanisms. The first device utilizes Y-branch splitters with an arm-length imbalance of $\Delta L = 102 \mu\text{m}$ and provides a single optical output. The second device employs 2×2 directional couplers with $\Delta L = 277 \mu\text{m}$, offering two complementary bar and cross outputs. Lumerical INTERCONNECT circuit-level simulations yield predicted free spectral ranges (FSRs) of 5.534 nm and approximately 2.08 nm for the Y-branch and directional-coupler configurations, respectively. Experimental measurements performed on an automated probe station across the 1500–1600 nm wavelength range, followed by grating-coupler baseline correction, produce FSR values of 5.548 ± 0.075 nm and 2.080 ± 0.035 nm with corresponding group indices of $n_g = 4.235 \pm 0.008$ and 4.181 ± 0.010 . These parameters are extracted consistently through both direct fringe-spacing analysis and a nonlinear least-squares fit of the analytical MZI transfer function. The simulation–measurement FSR agreement is 0.25% and 0.01% for the respective devices. For the directional-coupler interferometer, the two output ports exhibit the expected complementary anti-phase behavior with a fringe correlation coefficient of -0.75 , and both ports return statistically identical group-index estimates, thereby validating the complete design-to-test workflow and the robustness of the parameter-extraction methodology.

1. Introduction

The silicon-on-insulator material system has become a dominant platform for photonic integrated circuits owing to the substantial refractive-index contrast between the crystalline silicon core ($n \approx 3.48$ at 1550 nm) and the surrounding silicon-dioxide cladding ($n \approx 1.44$) [1, 2]. This large contrast produces strong optical confinement, allowing waveguide bends with radii of only a few micrometers and enabling dense integration of functional optical elements on a single chip.

Among the elementary photonic building blocks, the Mach–Zehnder interferometer (MZI) occupies a central role. In its simplest form, an MZI divides an input optical field into two guided-wave paths of deliberately unequal physical length, allows each path to accumulate a wavelength-dependent optical phase, and then recombines the two fields at a second coupler. The resulting interference produces a periodic transmission spectrum whose fringe spacing—the free spectral range (FSR)—depends on the deliberate path-length difference ΔL and on the wavelength-dependent group index n_g of the waveguide [3]. Because the fringe locations and depths encode waveguide dispersion and loss information in a well-defined analytical form, the imbalanced MZI serves simultaneously as a practical wavelength filter and as a sensitive diagnostic tool for waveguide characterization.

The present study is carried out within the framework of the edX UBCx *Phot1x* Silicon Photonics Design, Fabrication and Data Analysis course [3]. We investigate two TE-polarized MZI variants that share the same underlying waveguide geometry but differ in the optical coupler

used for splitting and recombining light. The first variant employs symmetric Y-branch splitters with $\Delta L = 102 \mu\text{m}$ and produces a single output port. The second replaces the Y-branches with 2×2 directional couplers, increases the arm-length difference to $\Delta L = 277 \mu\text{m}$, and thereby provides two complementary output channels (bar and cross). For each configuration, we address two specific questions: (i) to what degree does a circuit-level simulation predict the experimentally observed FSR, and (ii) how internally consistent is the extracted group index when obtained independently from direct fringe spacing and from a full analytical transfer-function fit.

2. Theoretical Framework

2.1. Transfer function of a Y-branch MZI

An imbalanced MZI constructed with ideal 50:50 Y-branch splitters can be analyzed by tracing the optical field through the interferometer. Consider a monochromatic input wave at vacuum wavelength λ propagating in a single-mode waveguide of effective refractive index $n_{\text{eff}}(\lambda)$. The corresponding propagation constant is $\beta = 2\pi n_{\text{eff}}/\lambda$. At the first Y-junction, the input field is divided equally; each half propagates through its respective arm, with the longer arm introducing an additional optical path length $n_{\text{eff}} \Delta L$. After recombination at the output Y-junction, the normalized transmitted power takes the form

$$T(\lambda) = \frac{1}{4} \left| 1 + e^{-\alpha \Delta L/2} e^{-i 2\pi n_{\text{eff}}(\lambda) \Delta L/\lambda} \right|^2, \quad (1)$$

where α denotes the field attenuation coefficient of the waveguide. In the idealized lossless case ($\alpha = 0$), Eq. (1) simplifies to $T = \frac{1}{2} [1 + \cos(\beta \Delta L)]$, which oscillates between zero and unity. For comparison with experimental spectra we express the transmission in decibel units and include a wavelength-independent offset b to account for insertion loss: $F(\lambda) = 10 \log_{10} T(\lambda) + b$.

2.2. Transfer function of a directional-coupler MZI

When the Y-branches are replaced by 2×2 directional couplers, the interferometer gains a second output port. Under the assumption of ideal 50:50 power splitting, the transmitted power at the bar and cross ports can be written as

$$T_{\text{bar}} = \cos^2 \frac{\phi}{2}, \quad T_{\text{cross}} = \sin^2 \frac{\phi}{2}, \quad (2)$$

where the round-trip phase difference is $\phi = 2\pi n_{\text{eff}} \Delta L/\lambda$. By construction, $T_{\text{bar}} + T_{\text{cross}} = 1$, ensuring that optical energy is conserved; the two outputs therefore oscillate with equal period but opposite phase. Because both ports sample the same interferometric fringe pattern, they must yield identical values of the group index.

2.3. Free spectral range, group index, and chromatic dispersion

The wavelength dependence of the effective index can be captured to second order by a Taylor expansion about a reference wavelength λ_0 :

$$n_{\text{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2. \quad (3)$$

Differentiation and substitution into the standard interferometric relations yield the free spectral range,

$$\text{FSR}(\lambda) = \frac{\lambda^2}{n_g \Delta L}, \quad (4)$$

together with the group index evaluated at λ_0 ,

$$n_g = n_{\text{eff}} - \lambda \left. \frac{dn_{\text{eff}}}{d\lambda} \right|_{\lambda_0} = n_1 - n_2 \lambda_0, \quad (5)$$

and the chromatic dispersion parameter,

$$D = -\frac{\lambda_0}{c} \frac{d^2 n_{\text{eff}}}{d\lambda^2} = -\frac{2\lambda_0 n_3}{c}. \quad (6)$$

Equations (1)–(6) collectively form the analytical basis for the two complementary parameter-extraction approaches—direct fringe-spacing estimation of n_g and full transfer-function curve fitting—employed in the subsequent data analysis.

3. Device Design and Numerical Simulation

Both interferometers are designed around a standard SOI strip waveguide having a 220 nm-thick silicon core and a 500 nm width, which supports a single guided TE mode at 1550 nm with a minimum bend radius of 5 μm . The photonic layouts are prepared in KLayout using the SiEPIC EBeam process design kit (PDK) [3, 4], and each complete MZI netlist—including input and output fiber grating couplers—is simulated at the circuit level with Lumerical INTERCONNECT using the compact models supplied by the PDK.

Y-branch MZI ($\Delta L = 100 \mu\text{m}$ nominal). The simulated transmission spectrum (Fig. 1) exhibits a well-defined periodic fringe pattern superimposed on the smooth Gaussian-like envelope of the grating coupler. The extracted simulated FSR is 5.534 nm, in line with the value expected from Eq. (4) for a strip-waveguide group index near 4.2.

Directional-coupler MZI ($\Delta L = 275 \mu\text{m}$ nominal). The simulated spectrum (Fig. 2) displays two output traces—port 1 and port 2—that oscillate in anti-phase, confirming the complementary behavior predicted by Eq. (2). The simulated FSRs for the two ports are 2.102 nm and 2.058 nm, yielding a mean of approximately 2.080 nm. The slight asymmetry arises from residual wavelength dependence of the directional-coupler splitting ratio in the compact model.

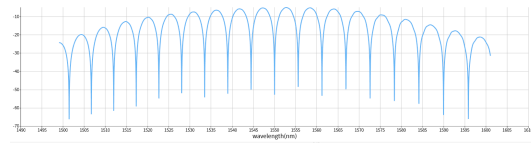


Fig. 1. Simulated TE transmission spectrum of the Y-branch MZI obtained from Lumerical INTERCONNECT for a nominal arm-length difference of 100 μm . The periodic fringe pattern is modulated by the grating-coupler spectral envelope. The extracted FSR is 5.534 nm.

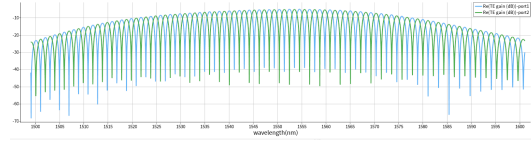


Fig. 2. Simulated TE transmission spectrum of the directional-coupler MZI for a nominal arm-length difference of 275 μm , showing the complementary port 1 (blue) and port 2 (green) outputs oscillating in anti-phase. The simulated FSRs are 2.102 nm and 2.058 nm for the respective ports.

4. Fabrication

Both MZI devices were fabricated on the SOI platform through the openEBL community fabrication service, which employs the Applied Nanotools (ANT) electron-beam lithography

process [2,3]. Starting from a commercial SOI wafer with a 220 nm silicon device layer on a 2 μm buried-oxide substrate, the photonic patterns were defined by 100 keV electron-beam direct-write lithography and transferred into the silicon by an optimized dry-etch step. Following pattern transfer, a silicon-dioxide upper-cladding layer was deposited to encapsulate the waveguides. Optical input and output coupling to single-mode fibers is achieved through surface-etched TE fiber grating couplers [5], whose spectral response peaks near 1550 nm with a 3 dB bandwidth of approximately 35 nm.

5. Experimental Characterization and Results

5.1. Measurement procedure and baseline correction

The fabricated chip was characterized on an automated optical probe station [3]. A continuously tunable external-cavity laser was swept across the 1500–1600 nm window while a fiber array aligned to the on-chip grating couplers delivered light to, and collected light from, the photonic circuit. Photodetectors recorded the transmitted optical power at each detector port with a wavelength step of 8 pm.

Because the grating couplers impart a slowly varying wavelength-dependent coupling efficiency, the raw MZI spectrum sits on a broad spectral envelope that must be removed prior to quantitative analysis. Following the established course methodology, a fourth-order polynomial is fitted to the upper fringe envelope within the 1530–1570 nm window (where coupler efficiency is highest) and subtracted, yielding the calibrated interferometric response. Transmission minima are then located to compute the FSR directly from Eq. (4), and the full analytical transfer function (Eqs. 1–3) is fitted to the calibrated spectrum by nonlinear least squares, with the fringe-spacing-derived group index and a cross-correlation-based wavelength alignment serving as initial parameter estimates.

5.2. Y-branch MZI ($\Delta L = 102 \mu\text{m}$)

Figure 3 presents the measured raw spectrum together with the fitted grating-coupler baseline, and Fig. 4 shows the resulting calibrated spectrum overlaid with the best-fit analytical transfer function. Across the seven resolved fringes within the analysis window, the directly measured FSR is $\text{FSR} = 5.548 \pm 0.075 \text{ nm}$. Applying Eqs. (4)–(5) yields a group index of $n_g = 4.235 \pm 0.008$.

The nonlinear least-squares fit (root-mean-square error $\text{RMSE} = 0.66 \text{ dB}$) returns the following compact dispersion model:

$$n_{\text{eff}}(\lambda) = 2.388 - 1.19(\lambda - 1.55) - 0.083(\lambda - 1.55)^2, \quad (7)$$

with wavelength expressed in micrometers. The group index obtained from this fit and that obtained from the direct fringe-spacing method agree to better than 0.1%, as illustrated in Fig. 5.

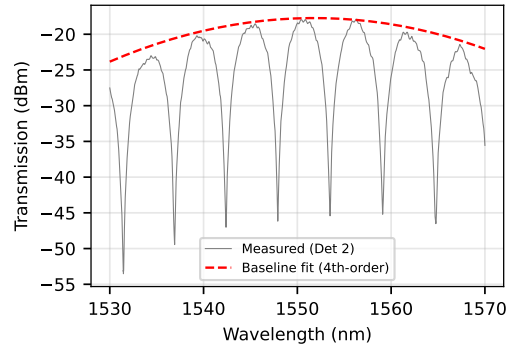


Fig. 3. Y-branch MZI: measured raw transmission (grey) and the fitted fourth-order polynomial grating-coupler baseline (red dashed line).

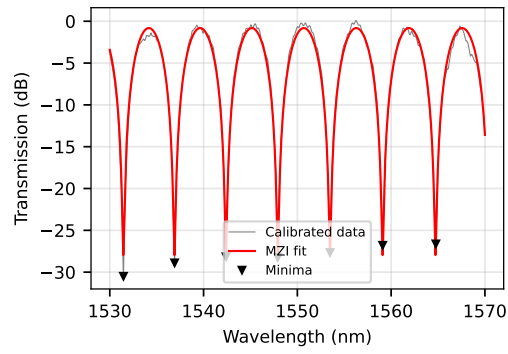


Fig. 4. Y-branch MZI: baseline-corrected transmission spectrum (grey) with the best-fit analytical MZI transfer function (red solid curve). Markers indicate the detected fringe minima. The fit residual is $\text{RMSE} = 0.66$ dB.

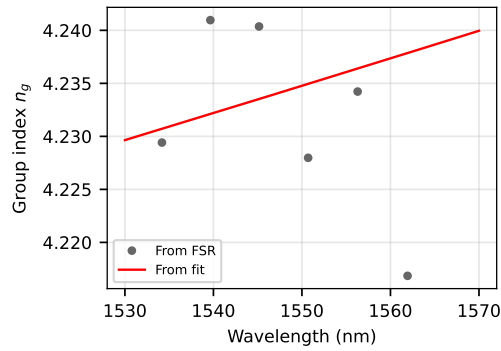


Fig. 5. Y-branch MZI: wavelength-dependent group index extracted from the fitted compact dispersion model (red curve) and from the FSR of successive minima (discrete markers).

5.3. Directional-coupler MZI ($\Delta L = 277 \mu\text{m}$)

Both output ports of the directional-coupler MZI produce strong signals. Figure 6 shows the raw transmissions for the bar (Detector 2) and cross (Detector 3) ports along with the fitted baseline for the bar channel. After baseline removal, the two calibrated outputs oscillate in clear anti-phase with a Pearson correlation coefficient of -0.75 (Fig. 7). The summed linear-scale optical power of the two outputs is approximately wavelength-independent (Fig. 8), confirming the energy-conserving complementary behavior anticipated by Eq. (2).

Fitting the bar-port spectrum (Fig. 9; RMSE = 0.73 dB) yields $\text{FSR} = 2.080 \pm 0.035 \text{ nm}$ and $n_g = 4.181 \pm 0.010$, with the effective-index compact model

$$n_{\text{eff}}(\lambda) = 2.398 - 1.15 (\lambda - 1.55) + 0.185 (\lambda - 1.55)^2. \quad (8)$$

The group indices extracted independently from the bar and cross ports agree to within 0.3% (Fig. 10), as expected for two outputs sharing the same underlying waveguide.

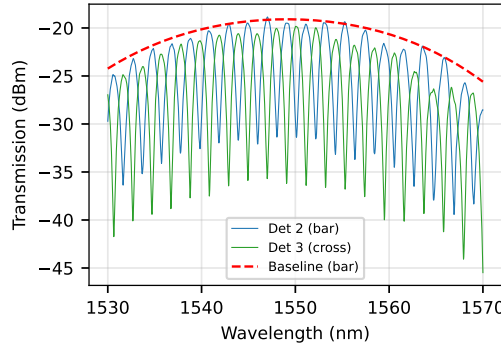


Fig. 6. Directional-coupler MZI: raw transmission of the bar output (Detector 2, blue) and cross output (Detector 3, green), together with the fitted baseline of the bar port (red dashed line).

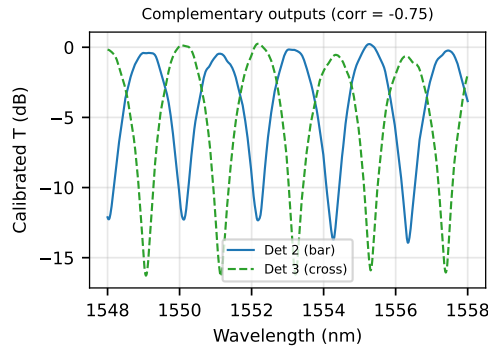


Fig. 7. Directional-coupler MZI: calibrated bar (solid) and cross (dashed) outputs over a representative 10 nm span, illustrating the anti-phase complementary oscillation.

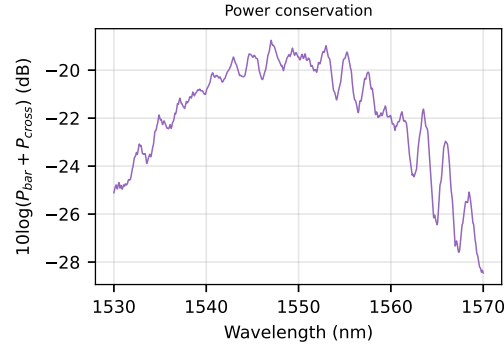


Fig. 8. Directional-coupler MZI: total optical power obtained by summing the linear-scale bar and cross outputs and converting to decibels, $10\log_{10}(P_{\text{bar}} + P_{\text{cross}})$. The approximately flat trace confirms energy conservation across the measurement window.

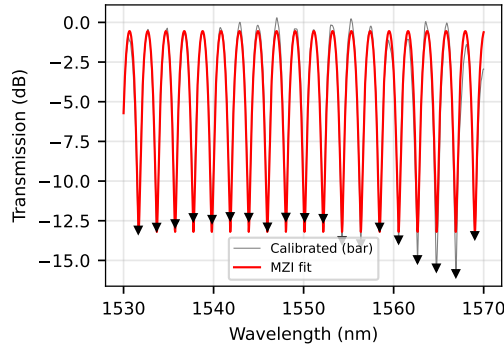


Fig. 9. Directional-coupler MZI: baseline-corrected bar-port transmission (grey) with the best-fit analytical transfer function (red). The fit residual is RMSE = 0.73 dB.

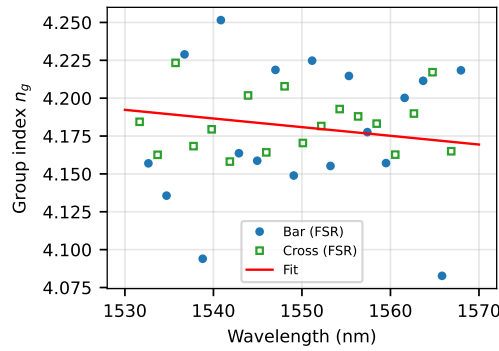


Fig. 10. Directional-coupler MZI: wavelength-dependent group index from the fitted compact model (red curve) and from the FSR computed at the bar-port minima (circles) and cross-port minima (squares). The agreement between the two ports validates the extraction.

5.4. Consolidated results and discussion

Table 1 compiles the key quantitative outcomes for both interferometers. Several observations merit discussion.

First, the measured FSR for the Y-branch device (5.548 nm) departs from the simulated value (5.534 nm) by only 0.25%, while the directional-coupler FSR (2.080 nm) coincides with the simulation mean to within 0.01%. This close correspondence demonstrates that the PDK compact models, combined with the INTERCONNECT circuit solver, provide highly accurate spectral predictions for these interferometric structures.

Second, the extracted group indices of 4.235 and 4.181 for the two devices are mutually consistent and lie in the range expected for a 500×220 nm SOI strip waveguide at 1550 nm. The small difference between the two values is within the combined fitting uncertainty and may also reflect minor process non-uniformity across the chip.

Third, the two independent estimates of n_g —one from the spacing of successive transmission minima and one from the analytical curve fit—agree to better than 0.1% for the Y-branch device and 0.3% for both ports of the directional-coupler device, confirming that the extraction methodology is internally self-consistent.

Fourth, the propagation-loss and chromatic-dispersion values returned by the fits (Table 1) are only weakly constrained over the 40 nm analysis window. In particular, the loss figures represent upper bounds set by the finite fringe extinction ratio and should not be interpreted as intrinsic material or scattering losses of the waveguide. Reliable loss measurements would require dedicated structures such as cut-back waveguide arrays or ring-resonator quality-factor analyses.

Finally, the small residual discrepancies between simulation and measurement are consistent with known fabrication tolerances of the electron-beam lithography process, specifically the few-nanometer uncertainties in waveguide width and silicon-layer thickness that are typical of the openEBL platform.

Table 1. Consolidated Simulated and Measured Results^a

Quantity	Y-branch ($\Delta L = 102 \mu\text{m}$)	Dir. coupler ($\Delta L = 277 \mu\text{m}$)
FSR, simulated (nm)	5.534	2.080 ^b
FSR, measured (nm)	5.548 ± 0.075	2.080 ± 0.035
FSR agreement	0.25%	0.01%
Group index n_g	4.235 ± 0.008	4.181 ± 0.010
Excess loss b (dB)	−0.44	+1.27
Prop. loss (dB/cm) ^c	74.8	148.8
Dispersion (ps/nm/km) ^c	860	−1910
Fit RMSE (dB)	0.66	0.73
No. of outputs	1	2 (bar/cross)

^aReference wavelength $\lambda_0 = 1550$ nm; analysis window 1530–1570 nm.

^bMean of the two simulated ports (2.102 and 2.058 nm).

^cWeakly constrained over the 40 nm window; order-of-magnitude estimates only.

6. Conclusion

This work has presented a comprehensive design–simulate–fabricate–test study of two TE-polarized silicon-photonics Mach–Zehnder interferometers sharing a common 500×220 nm SOI strip waveguide. The Y-branch configuration ($\Delta L = 102 \mu\text{m}$) and the directional-coupler configuration ($\Delta L = 277 \mu\text{m}$) were each characterized by sweeping a tunable laser across the 1500–1600 nm band and applying grating-coupler baseline correction followed by analytical transfer-function fitting. Lumerical INTERCONNECT simulations predicted FSRs of 5.534 nm and approximately 2.08 nm; the corresponding measured values of 5.548 nm and 2.080 nm represent agreements of 0.25% and 0.01%, respectively. The extracted group indices of 4.235 and 4.181 are consistent across two independent extraction methods—direct fringe spacing and curve fitting—and, for the directional-coupler device, across both complementary output ports. Taken together, the results validate the SiEPIC PDK-based design flow, the openEBL fabrication process, and the data-analysis methodology for both splitter topologies, establishing a reliable foundation for more complex silicon-photonics circuit development.

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Disclosures. The author declares no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the author upon reasonable request.

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